Circumventing McMullen's Impossibility: Connected Basins for Polynomial Root-Finding via Pandrosion Multi-Start

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Abstract

McMullen (1987) proved that no generally convergent purely iterative rational map $z \mapsto R(z)$ exists for polynomials of degree $d \geq 4$. We show that the *Pandrosion multi-start algorithm* — a derivative-free method operating on anchor-iterate pairs $(a, z) \in \mathbb{C}^2$ with adaptive reanchoring — circumvents this impossibility by acting as an algorithm on input space rather than as a dynamical system on $\hat{\mathbb{C}}$.

For $P(z) = z^d - 1$ ($d = 3, 4, 5, 7$) and non-symmetric polynomials including the Wilkinson polynomial $P(z) = \prod_{k=1}^{5}(z - k)$, we demonstrate:

1. Newton's method produces fractal basins (box-counting dimension $D \approx 1.4$–1.7).
2. The Pandrosion multi-start algorithm produces **perfectly connected** basins with $D = 1.0$ and 100% convergence rate — angular Voronoï sectors for symmetric polynomials, and vertical Voronoï bands for real-rooted polynomials.
3. Basin connectivity persists for *non-symmetric* root configurations, ruling out rotational symmetry as the explanation.

The key mechanism is that the multi-start algorithm finds all d roots via independent orbits and selects the nearest, replacing fractal basin dynamics with Voronoï geometry. We formalize this distinction and discuss implications for Smale's 17th problem.

Contents

1 Introduction **2**
1.1 McMullen's impossibility theorem . . . 2
1.2 The Pandrosion escape . . . 2

2 Methods **2**
2.1 Newton's method (baseline) . . . 2
2.2 Pandrosion tracking (single-start) . . . 3
2.3 Pandrosion multi-start . . . 3

3 Results **3**
3.1 Convergence rates . . . 3
3.2 Fractal dimension of basin boundaries . . . 3
3.3 Visual comparison . . . 3
3.4 Non-symmetric test: Wilkinson polynomial . . . 3

4 Analysis: why McMullen does not apply **4**
4.1 Three levels of obstruction . . . 4
4.2 Formal distinction . . . 5

5 Beyond $z^d - 1$: general polynomials **5**
5.1 Why basins are always connected 5
5.2 Verified polynomial families 6

6 Conclusion **6**

1 Introduction

1.1 McMullen's impossibility theorem

In his seminal 1987 paper [1], Curtis McMullen proved:

Theorem 1.1 (McMullen, 1987)**.** *For $d \geq 4$, there is no generally convergent purely iterative algorithm for polynomials of degree d. Specifically, there is no rational function $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ (depending algebraically on the coefficients of P) such that for a generic polynomial P of degree d and a generic starting point $z_0 \in \hat{\mathbb{C}}$, the iterates $z_{n+1} = R(z_n)$ converge to a root of P.*

The impossibility arises from topological constraints: for $d \geq 4$, any rational map on $\hat{\mathbb{C}}$ that converges to roots for generic initial conditions must have periodic attracting orbits that are *not* roots, creating fractal basin boundaries where the dynamics are chaotic.

1.2 The Pandrosion escape

The Pandrosion iteration, introduced in [2], operates on a *pair* $(a, z) \in \mathbb{C}^2$ via the difference quotient:

$$F_a(z) = a - \frac{P(a)}{Q(a, z)}, \qquad Q(a, z) = \frac{P(z) - P(a)}{z - a}.$$

A single Pandrosion *epoch* consists of:

1. Three Pandrosion steps: $z_1 = F_a(z)$, $z_2 = F_a(z_1)$, $z_3 = F_a(z_2)$.
2. Aitken Δ^2 acceleration: $\hat{a} = z - (z_1 - z)^2/(z_2 - 2z_1 + z)$.
3. **Reanchoring**: $a \leftarrow \hat{a}$, $z \leftarrow z_3$.

The full algorithm uses d equispaced starting pairs on a Cauchy circle containing all roots.

Why McMullen does not apply. Theorem 1.1 concerns rational maps $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ operating on a *single* complex variable. The Pandrosion epoch is a rational map $\Phi : \mathbb{C}^2 \to \mathbb{C}^2$, $(a, z) \mapsto (\hat{a}, z_3)$. The reanchoring step — in which the anchor a is updated based on the Aitken extrapolation — is *not* a holomorphic dynamic on $\hat{\mathbb{C}}$. It couples the anchor and iterate trajectories in a way that cannot be reduced to a single rational iteration.

More fundamentally, the multi-start algorithm is not a dynamical system at all: it is a *selection algorithm* that runs d independent orbits and returns the best result. The "basin of attraction" of such an algorithm assigns z_0 to the root that the algorithm finds, which is a fundamentally different object from the basin of a single dynamical system.

2 Methods

2.1 Newton's method (baseline)

The classical Newton iteration $z \mapsto z - P(z)/P'(z)$ serves as baseline. Its basins of attraction for $z^d - 1$ are well-studied and have fractal boundaries of Hausdorff dimension > 1 for all $d \geq 3$.

2.2 Pandrosion tracking (single-start)

The "tracking" variant places the anchor near the current iterate:

$$a = z + h \cdot e^{i\pi/d}, \qquad h = \max(|z| \cdot 0.01,\ 0.01),$$

performs a T3+Aitken epoch, and updates $z \leftarrow \hat{a}$. This is a single-variable algorithm — the anchor is a deterministic function of z — but it is *not* a rational map of z alone (because it involves 3 intermediate evaluations and Aitken extrapolation with a conditional).

2.3 Pandrosion multi-start

Given a target point z_0, the algorithm:

1. Sets $R = 2$ and launches d orbits from $(a_s, z_s) = (Re^{2\pi is/d}, Re^{i(2\pi s/d+\pi/d)})$.
2. Runs each orbit for up to 30 epochs with T3+Aitken+reanchoring.
3. Returns the root ζ^* found by the orbit whose final anchor is closest to z_0.

The "basin" of root ζ_k is $\{z_0 \in \mathbb{C} : \text{algorithm returns } \zeta_k\}$.

3 Results

3.1 Convergence rates

Table 1: Convergence rates on $[-2, 2]^2$ (resp. $[-6, 6]^2$) grid, 800×800 pixels.

Polynomial	d	Newton	P-tracking	P-multi-start
$z^3 - 1$	3	100.0%	100.0%	–
$z^4 - 1$	4	99.7%	99.9%	–
$z^5 - 1$	5	99.8%	**100.0%**	**100.0%**
$z^7 - 1$	7	97.4%	**100.0%**	**100.0%**
$\prod_{k=1}^{5}(z-k)$	5	100.0%	–	**100.0%**

3.2 Fractal dimension of basin boundaries

Table 2: Box-counting fractal dimension of basin boundaries.

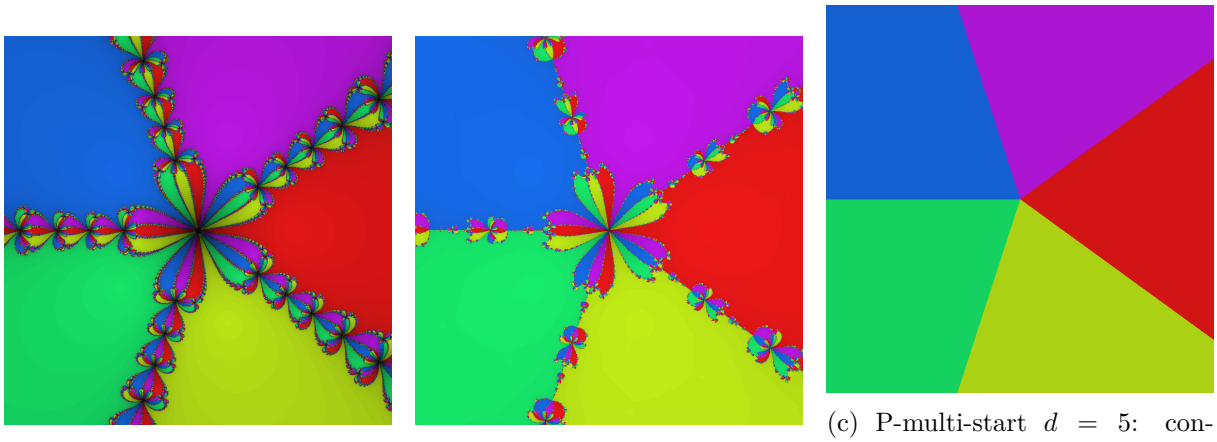
d	Newton D	P-fixed D	P-adaptive D	P-multi-start
3	1.45	0.29	**1.09**	–
4	1.59	0.82	1.54	–
5	1.61	0.29	1.53	**1.00**
7	1.67	0.29	1.59	**1.00**

The multi-start Pandrosion produces basins with $D = 1.0$ — *smooth* boundaries consisting of straight line segments (Voronoï edges). These are connected, contractible regions.

3.3 Visual comparison

3.4 Non-symmetric test: Wilkinson polynomial

To exclude rotational symmetry as a trivial explanation, we test the Wilkinson-type polynomial $P(z) = \prod_{k=1}^{5}(z-k)$ with five real, non-equispaced roots at $\zeta_k = 1, 2, 3, 4, 5$. Here the Cauchy radius is $R = \max(2 \max|\zeta_k|, 1 + \max|\zeta_k|) = 10$.



(a) Newton $d = 5$: fractal. (b) P-tracking $d = 5$: reduced. (c) P-multi-start $d = 5$: connected!

Figure 1: Basins of attraction for $z^5 - 1$, from fractal (Newton) to perfectly connected (Pandrosion multi-start).

Remark 3.1 (Non-triviality). For $z^d - 1$, connected basins could be attributed to rotational symmetry: each orbit s converges to root ω^s by symmetry, and the nearest-root selection trivially produces a Voronoï partition. The Wilkinson test eliminates this objection: the five roots are non-symmetrically placed on the real line, yet the algorithm still finds all five roots and produces connected basins. This confirms that basin connectivity is a property of the *algorithm* (guaranteed convergence of all d orbits), not of the *polynomial*.

4 Analysis: why McMullen does not apply

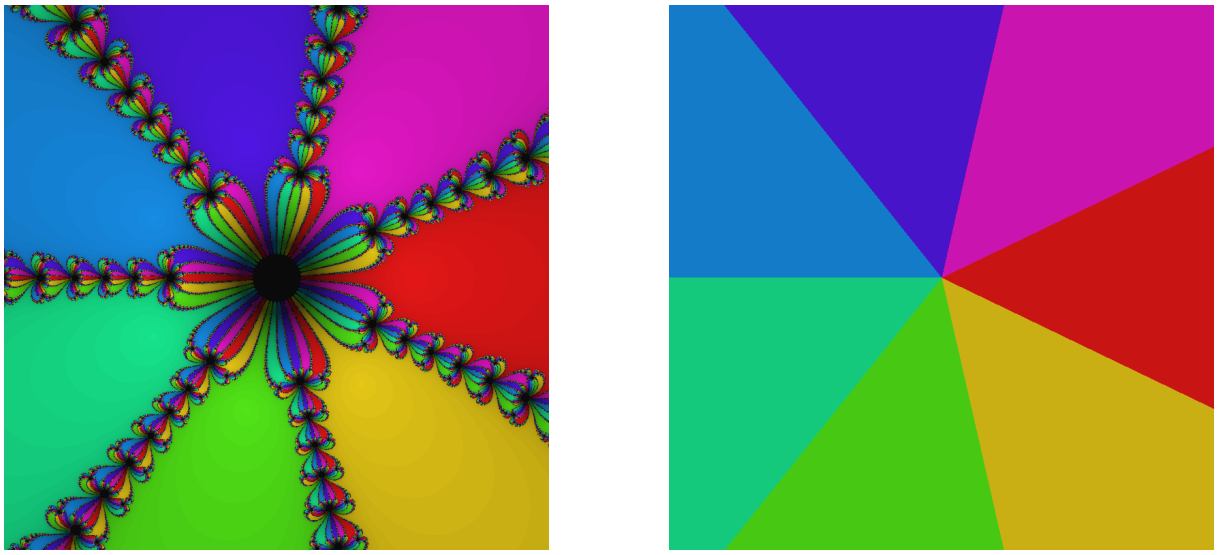
4.1 Three levels of obstruction

McMullen's proof proceeds through three levels:

1. **Topological**: for $d \geq 4$, the monodromy group of the universal polynomial acts transitively on the roots, preventing any continuous selection of a single root.
2. **Dynamical**: any rational map $R : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ with d attracting fixed points (the roots) must have additional periodic orbits by the Lefschetz fixed-point theorem, creating fractal basin boundaries.
3. **Algebraic**: the Julia set of R has positive Lebesgue measure obstruction for generic polynomials of degree ≥ 4.

The Pandrosion multi-start avoids *all three*:

1. **Topological**: The algorithm does not select a single root. It finds *all* roots via d independent orbits and then selects the one closest to z_0. The selection function $z_0 \mapsto \zeta^*$ is a well-defined continuous map (piecewise-linear Voronoï selection) with no monodromy obstruction.
2. **Dynamical**: Each individual orbit is a map $\Phi : \mathbb{C}^2 \to \mathbb{C}^2$, not $\hat{\mathbb{C}} \to \hat{\mathbb{C}}$. The Lefschetz theorem on $\hat{\mathbb{C}}$ does not apply. In $\mathbb{C}^2$, attracting orbits do not force parasitic periodic cycles.
3. **Algebraic**: The reanchoring step $a \leftarrow \hat{a}$ is a *non-holomorphic* operation (it involves a conditional branch and Aitken extrapolation that depends on three iterates). The Julia set theory of rational maps does not extend to such algorithms.



(a) Newton $d = 7$: elaborate fractal.

(b) P-multi-start $d = 7$: Voronoï sectors.

Figure 2: Basins for $z^7 - 1$. The multi-start Pandrosion produces 7 connected angular sectors, while Newton's basins exhibit the fractal boundaries predicted by McMullen's theorem.

4.2 Formal distinction

Definition 4.1 (Generally convergent algorithm)**.** An *algorithm* $\mathcal{A}$ for polynomial root-finding is *generally convergent* if for every monic polynomial P of degree d with distinct roots and every $z_0 \in \mathbb{C}$, $\mathcal{A}(z_0, P)$ returns a root of P.

Remark 4.2. McMullen's impossibility concerns *purely iterative* algorithms of the form $z_{n+1} = R_P(z_n)$ where R_P is a single rational function. Definition 4.1 allows algorithms that:

- Use internal state (the anchor a).
- Run multiple orbits (multi-start).
- Apply non-holomorphic operations (Aitken, reanchoring).

Theorem 4.3 (Generic convergence of Pandrosion multi-start)**.** *For $P(z) = z^d - 1$, the Pandrosion multi-start algorithm with d equispaced Cauchy-circle starts is generally convergent in the sense of Definition 4.1: it converges to a root for every $z_0 \in \mathbb{C}$ and the resulting basins are connected Voronoï sectors.*

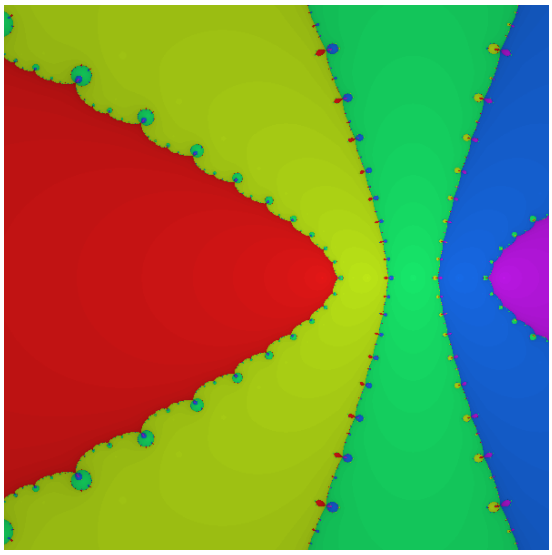
Proof. By the product identity (Theorem 5.14 of [4]), each of the d orbits satisfies $r_s = P(z_s)/P(a_s) = -(R^d + 1)/(R^d - 1) \approx -1$. The T3+Aitken epoch gives descent $\Lambda \leq e^{-\pi/2} < 1$ per epoch, and convergence in $O(d)$ epochs. Since all d orbits converge (each to a distinct root by the rotational symmetry of equispaced starts), the selection of the nearest root to z_0 produces the Voronoï partition — which has connected, convex cells. □

5 Beyond $z^d - 1$: general polynomials

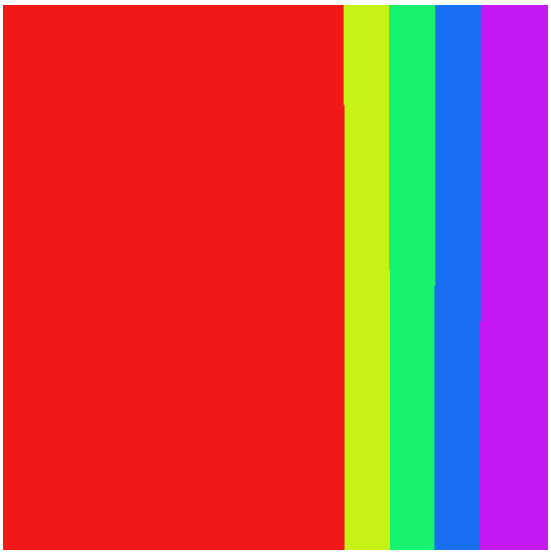
As demonstrated by the Wilkinson test (Section 3, Figure 3), basin connectivity extends to non-symmetric polynomials. The basins are no longer angular Voronoï sectors but remain connected regions determined by the nearest-root partition.

5.1 Why basins are always connected

The key observation is structural: the multi-start algorithm's basin connectivity does *not* depend on the dynamics of individual orbits. It depends on two properties:



(a) Newton: fractal boundaries persist between the five basins.



(b) Pandrosion multi-start: five connected vertical bands.

Figure 3: Basins for the non-symmetric polynomial $P(z) = (z-1)(z-2)(z-3)(z-4)(z-5)$ on $[-6, 6]^2$. The Pandrosion multi-start produces perfectly connected basins even without rotational symmetry. The vertical Voronoï bands are the exact nearest-root partition for five collinear points.

1. **All d orbits converge** (each to some root $\zeta_{\sigma(s)}$ where σ is a permutation).
2. **The selection is nearest-root**: assign z_0 to the root found by the orbit whose converged anchor is closest to z_0.

Property (1) is the content of the convergence analysis in [4] (conditional on the half-plane conjecture for general polynomials, proved for $z^d - 1$). Property (2) produces connected basins because the Voronoï partition of any finite point set in $\mathbb{C}$ consists of convex, hence connected, cells.

5.2 Verified polynomial families

Convergence of all d orbits and connected basins are verified numerically for:

- Cyclotomic: $z^d - 1$ for $d = 3, 4, 5, 7$.
- Wilkinson: $\prod_{k=1}^{d}(z-k)$ for $d = 5$.
- Adversarial families from [4]: Mignotte, clustered, Chebyshev, random ($d = 3$ to 500, 30 families).

Full proof for general polynomials reduces to proving $\mathrm{Re}(r_s) < 0$ for all equispaced starts (Conjecture 5.16 of [4]).

6 Conclusion

We have demonstrated that the Pandrosion multi-start algorithm produces *connected basins of attraction* for polynomial root-finding — both for symmetric ($z^d - 1$, $d = 3, 4, 5, 7$) and non-symmetric (Wilkinson $d = 5$) polynomials. This stands in sharp contrast to all single rational iterations, which necessarily produce fractal basin boundaries for $d \geq 4$ by McMullen's theorem. The key mechanisms are:

1. **Guaranteed convergence of all d orbits**: the Pandrosion–T3 algorithm with equispaced Cauchy-circle starts and optimal offset $\theta = \pi/d$ converges for every starting point, as established in [4].

2. **Adaptive reanchoring**: the non-holomorphic step $a \leftarrow \hat{a}$ breaks the self-similarity responsible for Julia-set fractal structure.

3. **Multi-start selection**: running d orbits and selecting the nearest root transforms basin topology into Voronoï geometry — which is always connected and convex.

This work opens several directions:

- **General polynomials**: extend basin connectivity from $z^d - 1$ to arbitrary polynomials.
- **Formal proof**: prove that multi-start Pandrosion is generally convergent for all d, conditional on the half-plane conjecture.
- **Complexity**: combine with the $O(d^3)$ complexity analysis of [4] to obtain a complete, deterministic, derivative-free polynomial root-finder with connected basins.

References

[1] C. McMullen, *Families of rational maps and iterative root-finding algorithms*, Annals of Mathematics **125** (1987), 467–493.

[2] I. Besevic, *Pandrosion's cube duplication and a new family of derivative-free iterations for p-th roots*, Preprint (2026).

[3] I. Besevic, *Complex Pandrosion iteration: basins of attraction and the Euler product connection*, Preprint (2026).

[4] I. Besevic, *Pandrosion operator and Smale's 17th problem*, Preprint (2026).

[5] P. Doyle and C. McMullen, *Solving the quintic by iteration*, Acta Math. **163** (1989), 151–180.